

# Index of Refraction

We must find the index of refraction of our material, polyethylene.

# 1 Theory

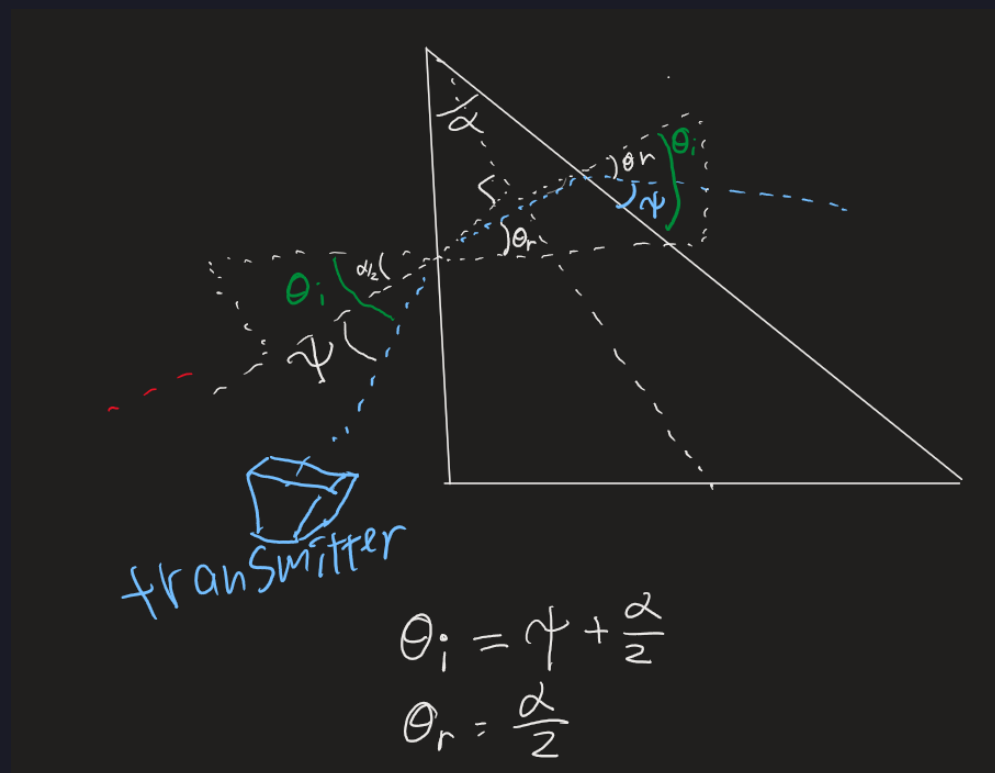
The index of refraction of our material can be found using Snell's law.

$$\frac{n_{\text{poly}}}{n_{\text{air}}} \equiv \frac{\sin \theta_i}{\sin \theta_r} \quad (1)$$

where  $\theta_i$  is the incident angle and  $\sin \theta_r$  is the reflected angle.  $N_{air} \approx 1$ .

In the diagram below, we position our transmitter at the left position at angle  $\psi$ , and our receiver on the right side at angle  $\psi$ . The angle that has the maximum transmission will read the largest value.

We can rewrite this definition in terms of experimental parameters:



With geometry, we can define an angle  $\psi$  off of  $\frac{\alpha}{2}$ . We use the line perpendicular to  $\frac{\alpha}{2}$  as our reference for 0 deg  $\phi$ , as this way a small error in our angle is relatively smaller, as long as we know  $\frac{\alpha}{2}$  to high precision. In our setup,  $\alpha \approx \frac{\pi}{2}$ .

Therefore, we have

$$n \equiv \frac{\sin\left(\psi + \frac{\pi}{4}\right)}{\sin\left(\frac{\pi}{4}\right)} \quad (2)$$

We can now measure our maximum transmission angle of incidence and use it to define our  $n$  for the material.

## 2 Experiment

We position our transmitter and receiver at incremental  $\psi$  to find the rough location where we get the greatest transmission. See Table 3 for a summary of the results.

We found the greatest intensity at  $\theta = 13$  degrees, which gives us:

$$n = 1.52 \pm 0.03$$

Table 3:

| Index of Refraction | 10x reading      |       |
|---------------------|------------------|-------|
| Psi                 | Receiver Reading | Error |
| 0                   | 0.1              | 0.025 |
| 5                   | 0.37             | 0.025 |
| 10                  | 0.7              | 0.025 |
| 15                  | 0.82             | 0.025 |
| 20                  | 0.75             | 0.025 |
| 17                  | 0.8              | 0.025 |
| 16                  | 0.81             | 0.025 |
| 14                  | 0.85             | 0.025 |
| 13                  | 0.88             | 0.025 |
| 12                  | 0.82             | 0.025 |